

C3 Parametric equations A level only

Name: _____

Class: _____

Date: _____

Time: **181 minutes**

Marks: **151 marks**

Comments:

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Q1.

A curve is defined by the parametric equations

$$x = 4 \times 2^{-t} + 3$$

$$y = 3 \times 2^t - 5$$

(a) Show that $\frac{dy}{dx} = -\frac{3}{4} \times 2^{2t}$

(3)

- (b) Find the Cartesian equation of the curve in the form $xy + ax + by = c$, where a , b and c are integers.

(3)

(Total 6 marks)

Q2.

A curve is defined by the parametric equations

$$x = t^3 + 2, \quad y = t^2 - 1$$

- (a) Find the gradient of the curve at the point where $t = -2$

(4)

- (b) Find a Cartesian equation of the curve.

(2)

(Total 6 marks)

Q3.

- (a) A curve is defined by the equation $x^2 - y^2 = 8$.

- (i) Show that at any point (p, q) on the curve, where $q \neq 0$, the gradient of the

curve is given by $\frac{dy}{dx} = \frac{p}{q}$.

(2)

- (ii) Show that the tangents at the points (p, q) and $(p, -q)$ intersect on the x -axis.

(4)

- (b) Show that $x = t + \frac{2}{t}$, $y = t - \frac{2}{t}$ are parametric equations of the curve $x^2 - y^2 = 8$.

(2)

(Total 8 marks)

Q4.

A curve is defined by the parametric equations $x = 8e^{-2t} - 4$, $y = 2e^{2t} + 4$.

- (a) Find $\frac{dy}{dx}$ in terms of t . (3)
- (b) The point P , where $t = \ln 2$, lies on the curve.
- (i) Find the gradient of the curve at P . (1)
- (ii) Find the coordinates of P . (2)
- (iii) The normal at P crosses the x -axis at the point Q . Find the coordinates of Q . (3)
- (c) Find the Cartesian equation of the curve in the form $xy + 4y - 4x = k$, where k is an integer. (3)
- (Total 12 marks)

Q5.

A curve is defined by the parametric equations

$$x = 8t^2 - t, \quad y = \frac{3}{t}$$

- (a) Show that the cartesian equation of the curve can be written as $xy^2 + 3y = k$, stating the value of the integer k . (2)
- (b) (i) Find an equation of the tangent to the curve at the point P , where $t = \frac{1}{4}$. (7)
- (ii) Verify that the tangent at P intersects the curve when $x = \frac{3}{2}$. (2)
- (Total 11 marks)

Q6.

A curve is defined by the parametric equations

$$x = 2 \cos \theta, \quad y = 3 \sin 2\theta$$

- (a) (i) Show that

$$\frac{dy}{dx} = a \sin \theta + b \operatorname{cosec} \theta$$

where a and b are integers.

(4)

- (ii) Find the gradient of the normal to the curve at the point where $\theta = \frac{\pi}{6}$.

(2)

- (b) Show that the cartesian equation of the curve can be expressed as

$$y^2 = px^2(4 - x^2)$$

where p is a rational number.

(3)

(Total 9 marks)

Q7.

A curve is defined by the parametric equations

$$x = 3e^t, \quad y = e^{2t} - e^{-2t}$$

- (a) (i) Find the gradient of the curve at the point where $t = 0$.
- (ii) Find an equation of the tangent to the curve at the point where $t = 0$.
- (b) Show that the cartesian equation of the curve can be written in the form

$$y = \frac{x^2}{k} - \frac{k}{x^2}$$

where k is an integer.

(2)

(Total 6 marks)

Q8.

- (a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

- (i) Show that $\frac{dy}{dx} = \frac{1}{k \cos \theta}$, where k is an integer.

(4)

- (ii) Find an equation of the normal to the curve at the point where $\theta = \frac{\pi}{3}$.

(4)

- (b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$.

(5)

(Total 13 marks)

Q9.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$.

(2)

- (ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$.

(2)

- (b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ .

(3)

- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P .

(3)

(Total 10 marks)

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Q10.

A curve is defined by the parametric equations

$$x = 1 - 3t, \quad y = 1 + 2t^3$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(3)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Find a cartesian equation of the curve.

(2)

(Total 9 marks)

Q11.

A curve is defined by the parametric equations

$$x = \frac{1}{t}, \quad y = t + \frac{1}{2t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t . (4)
- (b) Find an equation of the normal to the curve at the point where $t = 1$. (4)
- (c) Show that the cartesian equation of the curve can be written in the form

$$x^2 - 2xy + k = 0$$

where k is an integer.

(3)

(Total 11 marks)

Q12.

A curve is defined by the parametric equations $x = 2t + \frac{1}{t^2}, y = 2t - \frac{1}{t^2}$.

- (a) At the point P on the curve, $t = \frac{1}{2}$.
- (i) Find the coordinates of P . (2)
- (ii) Find an equation of the tangent to the curve at P . (5)
- (b) Show that the cartesian equation of the curve can be written as

$$(x - y)(x + y)^2 = k$$

where k is an integer.

(3)

(Total 10 marks)

Q13.

A curve is defined by the parametric equations

$$x = 1 + 2t, \quad y = 1 - 4t^2$$

- (a) (i) Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (2)
- (ii) Hence find $\frac{dy}{dx}$ in terms of t . (2)
- (b) Find an equation of the normal to the curve at the point where $t = 1$. (4)
- (c) Find a cartesian equation of the curve. (3)
- (Total 11 marks)**

Q14.

A curve is given by the parametric equations

$$x = \cos \theta \quad y = \sin 2\theta$$

- (a) (i) Find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$. (2)
- (ii) Find the gradient of the curve at the point where $\theta = \frac{\pi}{6}$. (2)
- (b) Show that the cartesian equation of the curve can be written as $y^2 = kx^2(1 - x^2)$ where k is an integer. (4)
- (Total 8 marks)**

Q15.

A curve is defined by the parametric equations

$$x = 3 - 4t \quad y = 1 + \frac{2}{t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t . (4)
- (b) Find the equation of the tangent to the curve at the point where $t = 2$, giving your

answer in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

- (c) Verify that the cartesian equation of the curve can be written as

$$(x - 3)(y - 1) + 8 = 0$$

(3)

(Total 11 marks)

Q16.

A curve is defined by the parametric equations

$$x = 2t + \frac{1}{t}, \quad y = \frac{1}{t}, \quad t \neq 0$$

- (a) Find the coordinates of the point on the curve where $t = \frac{1}{2}$.

(2)

- (b) Show that the cartesian equation of the curve can be written as

$$xy - y^2 = 2$$

(2)

- (c) Show that the gradient of the curve at the point (3, 2) is 2.

(6)

(Total 10 marks)